Ihor Kendiukhov
Institute of Medical Genetics and Applied Genomics
University of Tübingen, Tübingen, Germany
ihor.kendiukhov@student.uni-tuebingen.de

To the Editors, *Scientific Reports*
Submission ID: 43fb9b81-0ebd-4a9a-a390-5ba988a88425

Dear Dr Li,

Thank you for the opportunity to revise the manuscript previously titled "Certified Symmetry Discovery: Confidence Sets for Lie-Algebra Generators from Noisy, Limited Data". I am submitting the revised version, now titled **"Certified symmetry discovery for polynomial and dictionary vector fields: confidence sets for Lie-algebra generators from noisy vector-field measurements"**, together with a point-by-point response to the three reviewers and a separate Supplementary Information file.

The reviewers' central request, shared by the editorial assessment, was to align every practical and empirical claim with what the theorem actually guarantees. The revision does this through new theory, new code and new experiments rather than through rewording alone. In brief:

- a table listing each guarantee with the assumptions it needs, and a separate treatment of a fixed, a data-selected and a misspecified model class, with a new result quantifying how large a symmetry violation a lack-of-fit test of given power can miss and a new procedure that remains valid when the class is chosen from the data, provided the largest class considered contains the truth; a new simulation study evaluates both;
- the certificate is now the supremum of the defect over the confidence region, with the infimum as a lower bound, as one reviewer suggested, computed with certified numerics;
- dependence-robust (cluster and Newey–West) regions, with a new study of serially correlated errors that reports the coverage of the coefficient region itself;
- an output metric which, when transported with the coordinates, makes the defect, and hence the tolerance, invariant under every affine change of coordinates, a non-asymptotic resolution bound, exact binomial intervals for the worst-setting rates of the calibration table, for the zero false-certification rate of the certificate and for selected rates in the text, together with a stated bound on the Monte Carlo error of every other rate, a recovery metric, a bracket-closure diagnostic, an extension to non-polynomial dictionaries, a demonstration that noisy states can manufacture false symmetries, and a study showing that, in the systems tested, additive derivative noise only rarely makes a system without symmetry look symmetric, and then only near the tolerance, although it can tilt a real symmetry (the source of the threshold rule's 63% worst case);
- a second real data set, ocean drifter velocities in which rotation invariance is physically expected, analysed as a case study under stated assumptions;
- a redesigned tightness experiment, which also shows that a narrower bootstrap bound fails for a generator selected from the data while the certificate does not, the correction of an implementation error found while revisiting the trajectory study, and a narrower title.

All experiments were re-run with the final code, and the numerical results in the text are checked automatically against the result files. The manuscript uses the journal's structure: an unstructured abstract of at most 200 words, Introduction, Results, Discussion and Methods,

numbered references, eight display items, and the supporting material in a separate Supplementary Information file. The main text (Introduction, Results and Discussion, without figure legends) runs to about 12,500 words, and the reference list has 67 entries (59 of them cited in the main text, the rest only in Methods), above the recommended 4,500 words and about 60 references, mainly because of the analyses requested in review (data-driven class selection, dependence-robust inference, the tightness and noisy-state studies and the second real data set) and the limitations that accompany them. Proofs and secondary analyses are already in the Methods and the Supplementary Information, and I would be glad to move further material there if you prefer.

Declarations. This manuscript is original, has not been published previously, and is not under consideration by another journal. I am the sole author. I have no competing interests to declare. The work involves no human participants, no animals and no personal data. The implementation, raw results and scripts are released in the public repository `https://github.com/Kendiukhov/certified-symmetry-discovery`, where the version used for the revised manuscript will be tagged `v2.0.0`.

Thank you for considering the revised manuscript.

Yours sincerely,

Ihor Kendiukhov